Department of Mechanical and Production Engineering

Aarhus University

Varmetransmission

Author	Noah Rahbek Bigum Hansen	202405538
Course	Varmetransmission	290221U002
Instructor	Pourya Forooghi & Mugisa Y. Kano	
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Lecture 1: Introduction to Heat Transfer 2026-08-25

These are the notes for the obligatory Heat Transfer course on the 5th semester of the Bs.c. in Mechanical Engineering at Aarhus University taught by Pourya Forooghi and Mugisa Y. Kano.

1 Introduction to Heat Transfer

Heat transfer as a subject has already been treated in the Partial Differential Equations Course. Further, a related subject, namely Thermodynamics has also been studied. However, whereas Thermodynamics mainly concerns itself with questions such as what the eventual equilibrium temperature will be or the total amount of heat transferred, heat transfer as a subject is more interested in questions such as the time it takes or the rate of heat transfer.

Heat Transfer is used in both power generation, heating/cooling/refrigeration, the chemical industry, indoor climate, thermal management and much more. Further, new technology means heat transfer will remain very useful into the future as data centers, electronics, and turbine blades all need to meet higher cooling standards to maintain performance.

Before we start, a quick note on terminology is useful. We distinguish between *Heat*, *Heat transfer rate*, and *Heat flux*. Heat, $Q[\mathrm{J}]$, denotes the amount of heat transfered. The heat transfer rate, $\dot{Q}, [\mathrm{J\,s^{-1}}]$, denotes the heat Q transferred per unit time. And finally, Heat flux, $\dot{q}[\mathrm{W\,m^{-2}}]$, is the heat transfer rate $\dot{Q}$ per unit area.

1.1 Thermodynamics Recap

Here will be given a short Introduction to the most important concepts of Thermodynamics which we will be dealing with here. For a more in-depth review of the material see my notes on the subject.

The change in internal energy u and enthalpy $h = u + p/\rho$ with temperature is generally given as:

- Ideal Gasses:

$$\mathrm{d}u = c_v\,\mathrm{d}T$$
$$\mathrm{d}h = c_p\,\mathrm{d}T.$$

- Incompressible substances:

$$\mathrm{d}u = c\,\mathrm{d}T \quad (c \approx c_p \approx c_v).$$

Also, the 1st law of thermodynamics (the energy balance) states that the net energy transfer by heat, work, and mass from a system equals the change in the total energy of said system. This also implies that the rate of energy transfer equals the rate of change in total energy.

2 Heat Transfer Mechanisms

We mainly work with three different heat transfer mechanisms. These are *Conduction*, *Convection*, and *Radiation*.

2.1 Conduction

Conduction is the transfer of energy from more energetic particles of a substance to the adjacent less energetic ones. This happens in both solids and fluids, but always through a material medium, so it cannot happen in a vacuum.

The rate of conduction $\dot{Q}_{\text{cond}}$ is generally given on a form resembling:

$$\dot{Q}_{\text{cond}} \propto A\frac{\Delta T}{\Delta x}$$

where A is the conducting area, ΔT is the change in temperature and $\Delta x$is the thickness of the cross-section. This can be qualified by including the *thermal conductivity* $k = \left[\mathrm{W\,m^{-1}\,K^{-1}}\right]$ which gives:

$$\dot{Q}_{\text{cond}} = -kA\frac{\Delta T}{\Delta x}.$$

As $\Delta x \to 0$ this turns into *Fourier's law of conduction (1D)*, given as:

$$\dot{q}_{\text{cond}} = -k\frac{\mathrm{d}T}{\mathrm{d}x} \tag{1}$$

or in general vector form as

$$\dot{\boldsymbol{q}} = -k\nabla T.$$

The thermal conductivity varies with temperature.

We also define the *thermal diffusivity* α as

$$\alpha = \frac{k}{\rho c_p}.$$

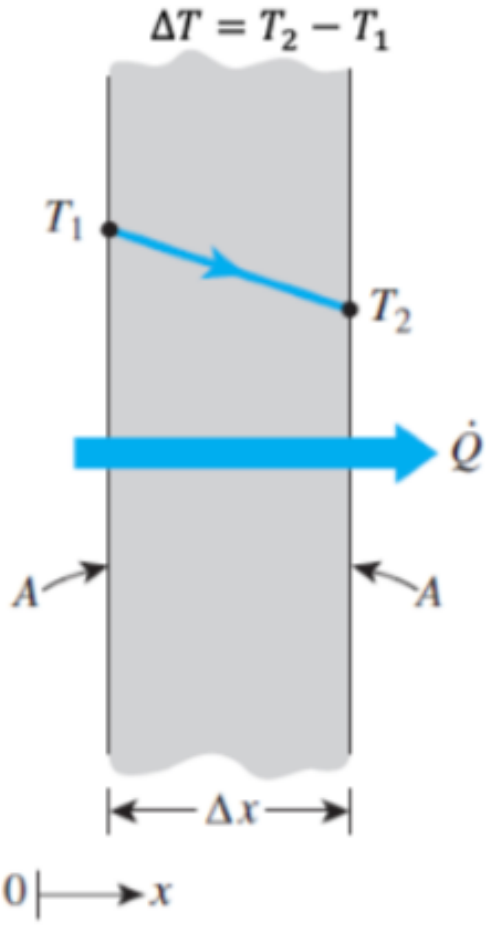


Figure 2.1: Cross section of a material displaying thermal conduction. Reproduced from (Çengel and Ghajar 2015, fig. 1–23).

Roughly speaking, the thermal diffusivity determines how fast heat diffuses in a certain solid material.

2.2 Convection

Convection is the transfer of energy as a result of bulk motion of fluid. This always happens between a solid surface and a flowing fluid. This can be both *natural (free) convection*, i.e. flow created by bouyancy force and *forced convection*, i.e. flow created by an external driving force. When a material undergoes convection it will generally also be undergoing convection to some degree.

The convection rate $\dot{Q}_{\text{conv}}$ is normally given in a form such as

$$\dot{Q}_{conv} =\propto A_s(T_s - T_\infty)$$

where T_∞ is the far field temperature (or some other reference temperature), T_s is the surface temperature, and A_s is the

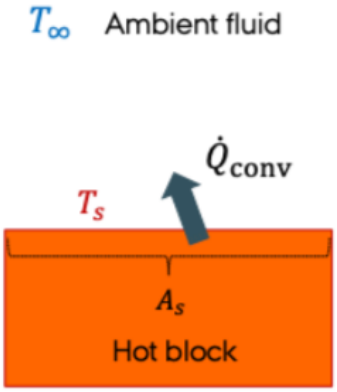


Figure 2.2: Cross section of a material displaying convection.

heat transfer surface area. By introducing the heat transfer coefficient $h\left[\mathrm{W\,m^{-2}\,K^{-1}}\right]$ we arrive at Newton's law of cooling:

$$\dot{Q}_{\mathrm{conv}} = hA_s\left(T_s - T_\infty\right). \tag{2}$$

It is important to remember the difference between the thermal conductivity k and the heat transfer coefficient h. Whereas the thermal conductivity is a material property the heat transfer coefficient is a property of the specific flow and must therefore be experimentally determined in all but the simplest cases.

2.3 Radiation

Thermal radiation is energy emitted by matter in form of electromagnetic waves. Here, there is no need for an intervening medium, so this may also happen in a vacuum. All bodies at a temperature above 0 K emit thermal radiation. In general, radiation is a volumetric phenomenon but for opaque solids it is usually considered a surface phenomenon. The heat transfer rate due to radiation, $\dot{Q}_{\mathrm{emit}}$:

$$\dot{Q}_{\mathrm{emit}} \propto A_s T_s^4.$$

For a black body (an ideal radiator) this is preciselu:

$$\dot{Q}_{\mathrm{emit}} = \sigma A_s T_s^4$$

where $\sigma = 5.670 \cdot 10^{-8}\,\mathrm{W/m^2/K^4}$ is Boltzmann's constant. The above is also known as Boltzmann's law. For an arbitrary surface we instead have

$$\dot{Q}_{\mathrm{emit}} = \epsilon\sigma A_s T_s^4$$

where $0 \leq \epsilon \leq 1$ is the emissivity.

Emissivity ϵ is a property of the radiating surface, which in reality varies with temperature but also depends on the wavelength of the electromagnetic wave. For simplicity, we often use its mean value.

The opposite case of emitted radiation is that of absorbed radiation given by

$$\dot{Q}_{\mathrm{abs}} = \alpha\dot{Q}_{\mathrm{incident}}$$

where $0 \leq \alpha \leq 1$ is the absorptivity. Kirchoff's law of radiation states that at the same temperature and wavelength $\epsilon = \alpha$.

The net radiation heat transfer of a solid surface is

$$\dot{Q}_{\mathrm{rad}} = \dot{Q}_{\mathrm{emit}} - \dot{Q}_{abs}.$$

We have an important special case of the above for a surface entirely enclosed by a much larger surrounding surface as shown on Figure 2.3, where

$$\dot{Q}_{\mathrm{rad}} = \epsilon\sigma A_s\left(T_s^4 - T_{\mathrm{surr}}^4\right).$$

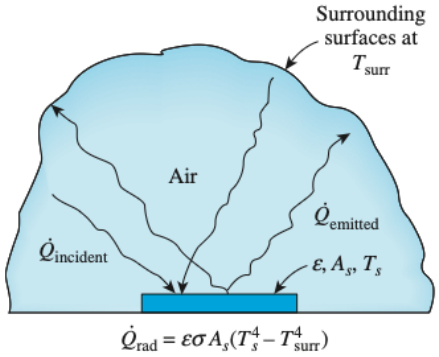


Figure 2.3: Radiation between enclosed and enclosing body. Reproduced from (Çengel and Ghajar 2015, fig. 1–40).

Lecture 2: Conduction, Pt. I 2026-09-01

3 Conduction

Heat transfer has both a *direction* and a *magnitude* and is therefore a vector quantity. The magnitude in a specified direction is proportional to the *temperature gradient*. Heat conduction in a medium is *steady*, when the temperature $T = T(x, y, z)$ does not vary with time and *unsteady* or *transient* when $T = (x, y, z, t)$ varies with time. During transient heat transfer, the temperature normally varies with time as well as position. In the special case of variation with time but not with position, the temperature of the medium changes uniformly with time – such systems are termed *lumped systems*. Many small objects can be analyzed as being lumped systems.

In Section 2.1, heat conduction was defined as the transfer of thermal energy from more energetic particles of a medium to the adjacent less energetic ones. We generally employ a coordinate system such that heat transfer in the positive direction of a coordinate axis is positive and vice versa.

We previously introduced Fourier's law of heat conduction in Equation (1),

$$\dot{Q}_{\text{cond}} = -kA\frac{\mathrm{d}T}{\mathrm{d}x}$$

where k is the *thermal conductivity* and $\frac{\mathrm{d}T}{\mathrm{d}x}$ is the temperature gradient. In general the thermal conductivity of a material k varies with time, however for most purposes an average value can be used for sufficiently accurate results.

Heat may also be generated throughout a medium. E.g. the temperature of a resistor rises volumetrically when a current is applied. Often it is also convenient to model the absorption of radiation such as solar energy or gamma rays as heat generation when those rays penetrate deep into a medium while being absorbed gradually such as for solar rays in a body of water. For opaque bodies almost all absorption occurs within the first few microns of the surface, so in these cases the solar energy can be treated as heat flux on the surface. As heat generation is volumetric, it is usually specified per unit volume and denoted by $\dot{e}_{\text{gen}} = \left[\mathrm{W\,m^{-3}}\right]$. The rate of heat generation may very with time as well as position. When the variation w.r.t. the position is

known, the total rate of heat generation in a medium of volume $\mathscr{V}$ can be determined from

$$\dot{E}_{\text{gen}} = \int_{\mathscr{V}} \dot{e}_{\text{gen}} \, \mathrm{d}\mathscr{V}. \tag{3}$$

In the special case of uniform heat generation Equation (3) reduces to

$$\dot{E}_{\text{gen}} = \dot{e}_{\text{gen}} \mathscr{V}.$$

3.1 Heat Conduction Equation

It is useful to expand Equation (1) to more dimensions and include the internal heat generation. In rectangular coordinates it can be shown that Equation (1) can be written as

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{e}_{\text{gen}} = \rho c \frac{\partial T}{\partial t}. \tag{4}$$

for density ρ and specific heat c. In cylindrical coordinates it can be written

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{e}_{\text{gen}} = \rho c \frac{\partial T}{\partial t}$$

and in spherical coordinates it is

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial}{\partial \theta}\left(k\sin\theta\frac{\partial T}{\partial \theta}\right) + \dot{e}_{\text{gen}} = \rho c \frac{\partial T}{\partial t}.$$

Equation (4) is the general heat conduction equation in rectangular coordinates. In the case of constant thermal conductivity it reduces to:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{e}_{\text{gen}}}{k} = \nabla^2 T + \frac{\dot{e}_{\text{gen}}}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t} \tag{5}$$

where $\alpha = k/\rho c$ is the *thermal diffusivity of the material*. Equation (5) is known as the Fourier-Biot Equation.

3.1.1 Special Cases of the Heat Equation

For steady state systems the temperature does not vary with time $\frac{\partial T}{\partial t} = 0$, so for steady-state:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{e}_{\text{gen}}}{k} = \nabla^2 T + \frac{\dot{e}_{\text{gen}}}{k} = 0. \tag{6}$$

Equation (6) is known as the Poisson equation.

For transient systems with no heat generation Equation (5) reduces to:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \nabla^2 T = \frac{1}{\alpha}\frac{\partial T}{\partial t}. \tag{7}$$

Equation (7) is also known as the diffusion equation.

For steady-state systems without heat generation we arrive at Section 3.1.1: Special Cases of the

Heat Equation also known as the Laplace equation:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \nabla^2 T = 0.$$

3.2 One-Dimensional Steady Heat Conduction

We consider heat conduction through a large plane wall such as the wall of a house, the glass of a single pane window, or many other cases. Heat conduction in these conditions can be approximated as being *one-dimensional* since the conduction is dominant in one direction and negligible in the others.

3.2.1 Plane Walls

We consider a thin element of thickness Δx in a large plane wall, as shown in Figure 3.1. We assume the density of the wall to be ρ, the specific heat to be c and the area of the wall normal to the direction of heat transfer to be A.

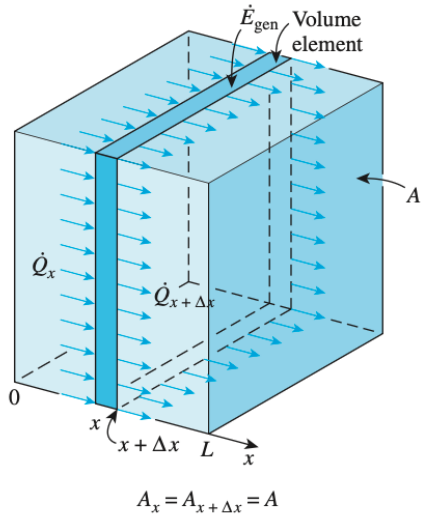


Figure 3.1: One-dimensional heat conduction through a volume element in a large plane wall. Reproduced from (Çengel and Ghajar 2015, fig 2–12).

Taking the direction normal to the surface of the wall, i.e. through the plane on Figure 3.1 to be the x-direction we obtain:

$$\begin{aligned}\frac{\mathrm{d}^2 T}{\mathrm{d}x^2} &= 0\\ \frac{\mathrm{d}T}{\mathrm{d}x} &= C_1\\ T &= C_1 x + C_2.\end{aligned}$$

On one side of the wall the temperature $T(0) = T_1$ and expressing the thickness of the wall as L the boundary condition on the other side of the wall is $T(L) = T_2$ as shown on Figure 3.2

As such

$$T(x) = T_1 + (T_2 - T_1)\frac{x}{L}.$$

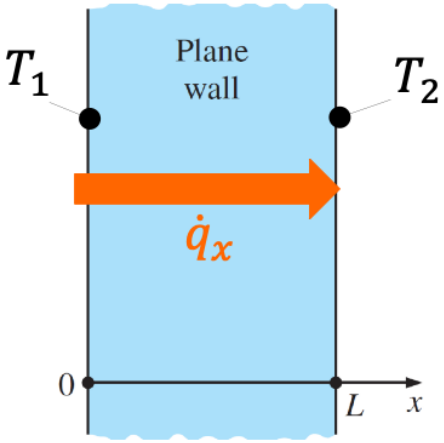


Figure 3.2: Conduction through a 1D-wall.

This can then be used in Equation (1) as

$$\begin{aligned} \dot{q}_x &= -k\frac{\mathrm{d}T}{\mathrm{d}x} \\ \dot{q}_x(x) = \text{const.} &= k\frac{T_1 - T_2}{L} \\ \dot{Q} &= k\frac{T_1 - T_2}{L}A. \end{aligned}$$

3.2.2 Cylindrical Geometries

We consider a pipe of length L, inner and outer radiusses r_1 and r_2 and thermal conductivity k as shown on Figure 3.3.

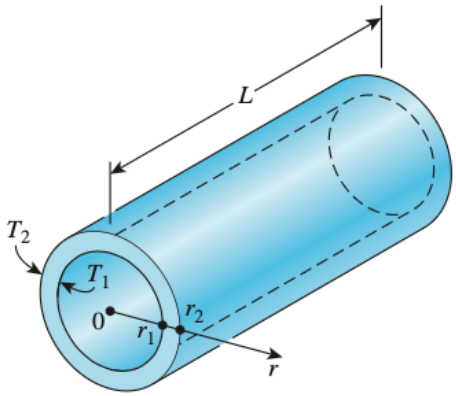


Figure 3.3: Schematic of a long cylinder with thickness $\Delta r = r_2 - r_1$. Reproduced from (Çengel and Ghajar 2015, fig. 2–50).

The governing differential equation in this case is

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}T}{\mathrm{d}r}\right) = 0$$

with boundary conditions $T(r_1) = T_1, T(r_2) = T_2$. Integrating the differential equation with respect

to r gives

$$r\frac{\mathrm{d}T}{\mathrm{d}r} = C_1 \frac{\mathrm{d}T}{\mathrm{d}r} = \frac{C_1}{r}$$
$$T(r) = C_1 \ln r + C_2.$$

Applying the boundary conditions this becomes

$$T(r) = T_1 + (T_2 - T_1)\frac{\ln(r/r_1)}{\ln(r_2/r_1)}.$$

This can be used in Equation (1) as

$$\dot{q}_r(r) = -k\frac{\mathrm{d}T}{\mathrm{d}r} = k\frac{T_1 - T_2}{\ln(r_2/r_1)r}$$
$$\dot{Q} = 2\pi lk\frac{T_1 - T_2}{\ln(r_2/r_1)}.$$

3.2.3 Internal Heat Generation

We now consider a long cylinder/wire which generates heat uniformly as shown on Figure 3.4. From steady state on cylindrical coordinates with uniform internal heat generatio we have

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}T}{\mathrm{d}r}\right) + \frac{\dot{e}_{\text{gen}}}{k} = 0$$
$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}T}{\mathrm{d}r}\right) = -\frac{\dot{e}_{\text{gen}}}{k}r$$
$$r\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{\dot{e}_{\text{gen}}}{k}\frac{r^2}{2} + C_1.$$

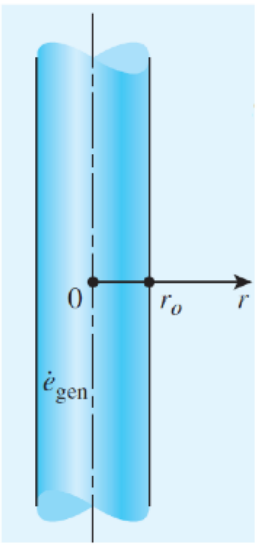


Figure 3.4: Cylinder/wire conducting heat with uniform internal heat generation.

Due to symmetry we have a boundary condition at $r = 0$ as

$$\frac{\mathrm{d}T}{\mathrm{d}r}(r = 0) = 0 \qquad \Longrightarrow \qquad C_1 = 0$$
$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{\dot{e}_{\text{gen}}}{k}\frac{r}{2}$$
$$T = -\frac{\dot{e}_{\text{gen}}}{k}\frac{r^2}{4} + C_2.$$

Our second boundary condition is that the temperature at the edge is the surface temperature $T(r_0) = T_s$. Hence

$$T = T_s + \frac{\dot{e}_{\text{gen}}}{4k}\left(r_0^2 - r^2\right)$$
$$T(r = 0) = T_s + \frac{\dot{e}_{\text{gen}}}{4k}r_0^2.$$

3.3 Boundary Conditions

A few different types of boundary conditions are normally employed

1. Specified Temperature (Dirichlet)
2. Specified heat flux (Neumann)
3. Insulated boundary – special case of Neumann with $\dot{q}_0 = 0$
4. Thermal symmetry – Symmetric temperature profile
5. Convection (Robin / mixed) – Specified convection from surfaces

3.4 Thermal Resistance

Under steady conditions and surface temperatures the cumbersome differential equations can be avoided by instead introducing the concept of *thermal resistivity* analogous to electrical resistivity.

3.4.1 Conduction Resistance

Analogous to electrical resistance:

$$I = \frac{V_1 - V_2}{R_e}$$

we now introduce thermal resistance

$$\dot{Q} = \frac{T_1 - T_2}{R}.$$

From the result for the conduction through a plane wall in Section 3.2.1: Plane Walls

$$\dot{Q} = k\frac{T_1 - T_2}{L}A$$

the thermal resistivity can by comparison with the definition be read to equal

$$R_{\text{cond, wall}} = \frac{L}{kA}.$$

From the result for the conduction through a cylinder from Section 3.2.2: Cylindrical Geometries

$$\dot{Q} = 2\pi lk\frac{T_1 - T_2}{\ln(r_2/r_1)}$$

we see that the thermal resistivity here is

$$R_{\text{cond,cyl}} = \frac{\ln(r_2/r_1)}{2\pi lk}.$$

3.4.2 Series Resistance Networks

The above conduction resistances can be combined. Say two walls are placed against each other. Following the above the thermal conduction through each wall is

$$\dot{Q}_2 = \frac{T_1 - T_2}{R_{\text{wall},1}}$$
$$\dot{Q}_3 = \frac{T_2 - T_3}{R_{\text{wall},2}}.$$

We can then eliminate the intermediate temperature at the interface between the walls T_2 as

$$\dot{Q}_{23} = \frac{T_1 - T_3}{R_{\text{wall},1} + R_{\text{wall},2}}.$$

We define the convective thermal resistance as

$$\dot{Q} = hA\left(T_\infty - T_s\right).$$

And as such for the case shown on Figure 3.5

$$\dot{Q}_1 = h_1 A\left(T_{\infty 1} - T_1\right) = \frac{T_{\infty 1} - T_1}{R_{\text{conv},1}}$$
$$\dot{Q}_4 = h_2 A\left(T_3 - T_{\infty 2}\right) = \frac{T_3 - T_{\infty 2}}{R_{\text{conv},2}}.$$

These can then all be combined to yield the total conduction through the wall

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{conv},1} + R_{\text{conv},2} + R_{\text{wall},1} + R_{\text{wall},2}}.$$

3.4.3 Parallel and Generalized Resistance Networks

Analogous to electrical resistances thermal resistances can also be in parallel as shown on Figure 3.6. In such cases the total resistance, R_{tot} is calculated using the reciprocal of the reciprocal sums of the individual parallel resistances.

$$\frac{1}{R_{\text{tot}}} = \frac{1}{R_1} + \frac{1}{R_2}.$$

Parallel thermal resistances only occur in 2D or 3D whilst thermal resistance theory is based on 1D solutions so this is only an approximation.

3.4.4 Thermal Contact Resistance

In the above, we assumed “perfect contact” at the interface between layers, and thus no temperature drop at the interface. In reality no such interface exists and as a result any interface will contain imperfections and air gaps. Thus, an interface offers some resistance to heat transfer which is characterised by a *thermal contact resistance* R_c for a unit interface area. This can be expressed in

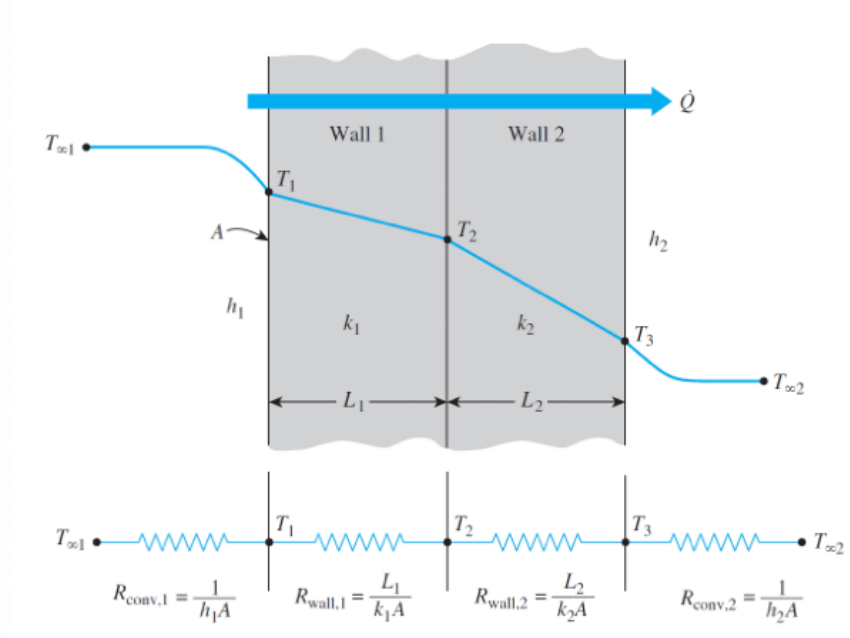


Figure 3.5: Thermal resistance in series.

a manner analogous to Newton's law of cooling

$$\dot{Q} = h_c A \Delta T_{\text{interface}}$$

where A is the apparent interface area, $\Delta T_{\text{interface}}$ is the effective temperature difference at the interface and h_c is the *thermal contact conductance* defined as

$$h_c = \frac{\dot{Q}/A}{\Delta T_{\text{interface}}}$$

which is the inverse of thermal contact resistance R_c

$$R_c = \frac{1}{h_c} = \frac{\Delta T_{\text{interface}}}{\dot{Q}/A}.$$

It is also worth noting that the heat transfer rate is the quotient of the temperature difference over the interface $\Delta T_{\text{interface}}$ and the created thermal resistance R_c/A as

$$\dot{Q} = \frac{\Delta T_{\text{interface}}}{R_c/A}.$$

3.5 Steady Heat Conduction in 2D & 3D

3.5.1 Analytical and Numerical Solutions

For multidimensional steady conduction, the governing equation is the steady heat conduction equation,

$$\nabla^2 T + \frac{\dot{e}_{\text{gen}}}{k} = 0..$$

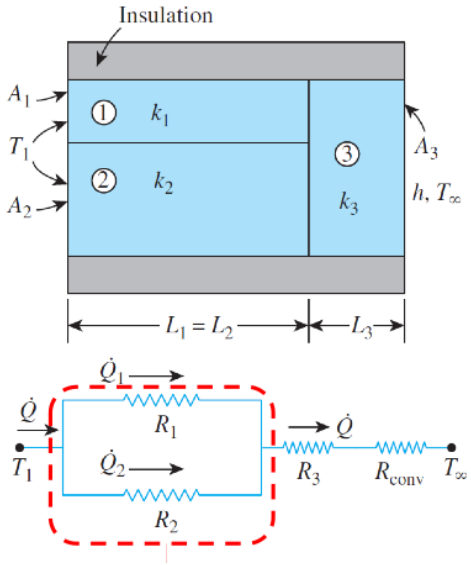


Figure 3.6: Parallel thermal resistance

Analytical solutions exist for relatively simple geometries and boundary conditions, but rapidly become cumbersome. Practical multidimensional problems are therefore commonly solved numerically or by use of existing analytical solutions. These existing analytical solution are often given in term of a shape factor S[m].

3.5.2 Conduction Shape Factors

The shape factor is defined such, that the stationary heat flow:

$$q''_x = -k\frac{\partial T}{\partial x}$$
$$q''_y = -k\frac{\partial T}{\partial y}$$

can be written on a simple form:

$$Q = Sk\Delta T_{1-2}. \tag{8}$$

Hence, the shape factors can only be used when considering conduction between two surfaces with constant temperature. The conduction shape factors depend only on the *geometry* of the system. These conduction shape factors S have been determined for a number of standard configurations and a small excerpt of these are shown in (Çengel and Ghajar 2015, Table 3–7). These are also only applicable when the mode of heat transfer is conduction and they are therefore not useful for liquids or gasses. By observation of Equation (8) it is evident that the conduction shape factor S is related to thermal resistance R by

$$R = \frac{1}{kS} \qquad \Longleftrightarrow \qquad S = \frac{1}{kR}.$$

3.6 Extended and Finned Surfaces

As mentioned previously, the rate of heat transfer from a surface at temperature T_s to the surrounding medium at temperature T_∞ is per Newton's law of cooling, Equation (2):

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty).$$

Hence, the heat transfer rate can be increased by either increasing the temperature difference, the convection heat transfer coefficient h or the surface area A_s. The temperature difference is often fixed by design constraints and increasing h may require installing a pump or fan or replacing an existing one with a larger one, but this may also not always be practical. The other alternative is increasing surface area by attaching *extended surfaces* called *fins* made of highly conductive material to the surface. The addition of fins often increase the heat transfer from a surface severalfold.

3.6.1 Fin Equation

From an energy balance on a fin, using Fourier's law of heat conduction it can be shown that the differential equation governing heat transfer in fins is:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(kA_c\frac{\mathrm{d}T}{\mathrm{d}x}\right) = hp(T - T_\infty).$$

In general, the cross sectional area A_c and the perimeter p of a fin vary with x which makes the above differential equation hard to solve. For *constant cross section* and *constant thermal conductivity* the above reduces to:

$$\frac{\mathrm{d}^2T}{\mathrm{d}x^2} = \frac{hp}{kA_c}(T - T_\infty).$$

This can also be expressed as

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}x^2} = m^2\theta \tag{9}$$

where

$$m^2 = \frac{hp}{kA_c}$$

and $\theta = T - T_\infty$ is the *temperature excess*. As m in Equation (9) is assumed constant the differential equation can be solved yielding

$$\theta(x) = C_1e^{mx} + C_2e^{-mx}$$

where C_1 and C_2 are arbitrary constants whose values depend on the boundary conditions at the base and at the tip of the fin.

The temperature of the plate to which the fin is attached is normally known in advance. Therefore, at the fin base we have a Dirichlet BC, expressed as

$$\theta(0) = \theta_b = T_b - T_\infty.$$

At the fin tip, however, we have several possibilities, which are explained in Section 3.6.2: Boundary Conditions for Fins.

3.6.2 Boundary Conditions for Fins

Infinitely Long Fins For a sufficiently long fin of uniform cross section, the temperature of the fin at the fin tip $T(L)$ approaches the environment temperature T_∞ and thus $\theta = T(L) - T_\infty \to 0$. Therefore, for an infinitely long fin we have the boundary condition at the fin tip:

$$\theta(L) = T(L) - T_\infty = 0, \qquad L \to \infty.$$

In this case our general solution after applying BC's become:

$$\theta(x) = \theta_b e^{-mx} = \theta_b e^{-x\sqrt{\frac{hp}{kA_c}}}.$$

This corresponds to the temperature decreasing exponentially from T_b to T_∞. The steady rate of heat transfer for an infinitely long fin can be determined from Fourier's law of heat conduction

$$\dot{Q}_{\text{long fin}} = -kA_c \frac{\mathrm{d}T}{\mathrm{d}x}\bigg|_{x=0} = \sqrt{hpkA_c}\theta_b$$

where p is the perimeter, A_c is the cross-sectional area of the fin, and x is the distance from the fin base.

Negligible Heat Loss from the Fin Tip Few fins are actually sufficiently long that their temperature approaches the surrounding temperature at the tip. A more realistic expectation is for heat transfer from the fin tip to be negligible since the heat transfer is proportional to surface area and little of the total surface area is in the tip. In such cases, the fin tip can be assumed adiabatic yielding the following boundary condition at the fin tip:

$$\frac{\mathrm{d}\theta}{\mathrm{d}x}\bigg|_{x=L} = 0.$$

After applying both BCs to the solution to the differential equation we get the following expression for the temperature excess along the fin:

$$\theta(x) = \theta_b \frac{\cosh m(L-x)}{\cosh mL}$$

and the rate of heat transfer can, again, be determined from Fourier's law of heat conduction:

$$\dot{Q}_{\text{adiabatic tip}} = -kA_c \frac{\mathrm{d}T}{\mathrm{d}x}\bigg|_{x=0} = \sqrt{hpkA_c}\theta_b \tanh mL.$$

Specified Temperature Instead of assuming the fin tip does not transfer heat or that the fin is sufficiently long for the fin tip to reach the surrounding temperature, it is often more accurate to specify a temperature at the fin tip which is assumed to be fixed. This temperature is often a bit higher than room temperature but the temperature excess at the tip under normal operating conditions must often be determined exponentially. In this case, the boundary condition for the fin tip is

$$\theta(L) = \theta_L = T_L - T_\infty.$$

In this case the temperature distribution along the wing is described by

$$\theta(x) = \theta_b \frac{\left(\frac{\theta_L}{\theta_b}\right)\sinh mx + \sinh m(L-x)}{\sinh mL}.$$

From Fourier's law of heat condution, the rate of heat transfer from the fin in this case is

$$\dot{Q}_{\text{specified temperature}} = -kA_c \frac{\mathrm{d}T}{\mathrm{d}x}\bigg|_{x=0} = \sqrt{hpkA_c}\,\theta_b \frac{\cosh mL - \frac{\theta_L}{\theta_b}}{\sinh mL}.$$

Convection from Fin Tip In actuality, fin tips are exposed to the surroundings, and thus the proper boundary conditions for the fin tip is convection that also may include the effects of radiation. In this case, the boundary condition at the fin tip can be obtained from an energy balance at the fin tip ($\dot{Q}_{\text{cond}} = \dot{Q}_{\text{conv}}$. This yields the boundary condition:

$$-kA_c \frac{\mathrm{d}T}{\mathrm{d}x}\bigg|_{x=L} = hA_c\theta_L.$$

In this case, the temperature profile across the fin is

$$\theta(x) = \theta_b \frac{\cosh m(L-x) + \left(\frac{h}{mk}\right)\sinh m(L-x)}{\cosh mL + \left(\frac{h}{mk}\right)\sinh mL}.$$

The rate of heat transfer here is

$$\dot{Q}_{\text{convection}} = -kA_c \frac{\mathrm{d}T}{\mathrm{d}x}\bigg|_{x=0} = \sqrt{hpkA_c}\,\theta_b \frac{\sinh mL + \frac{h}{mk}\cosh mL}{\cosh mL + \frac{h}{mk}\sinh mL}.$$

3.6.3 Fin Efficiency

We now consider the surface of a *plane wall* at temperature T_b exposed to a medium at temperature T_∞. Heat is lost from the surface to the surrounding medium by convection with a heat tranfer coefficient of h. Disregarding radiation, heat transfer from a surface area A_s is $\dot{Q} = hA_s(T_s - T_\infty)$.

Now we attach a fin of constant cross sectional-area A_c and length L to the surface with perfect contact. Now, heat is transferred from the surface to the fin by conduction and from the fin to the medium by convection with the same heat transfer coefficient h. The temperature of the fin is T_b at the base and gradually decreases toward the tip. Convection from the fin causes the temperature to drop somewhat from the midsection toward the outer surface. However, the cross-sectional area of the fin is usually very small and thus the temperature at any cross section can be assumed uniform.

In the limiting case of zero thermal resistance or infinite thermal conductivity, the temperature of the fin is uniform at the base at T_b. The heat transfer from the fin is maximum in this case and can be expressed as

$$\dot{Q}_{\text{fin, max}} = hA_{\text{fin}}(T_b - T_\infty).$$

In reality, the temperatyre of the fin drops along the fin and thus the heat transfer from the fin is less because of the decreasing temperature excess $T(x) - T_\infty$. To account for this, we define fin

efficiency as

$$\eta_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{fin, max}}} = \frac{\text{actual heat transfer rate from the fin}}{\text{Heat transfer rate from the fin if the entire fin was at base temperature}}.$$

Hence, the total heat transfer from a finned surface is

$$\dot{Q}_{\text{fin}} = \eta_{\text{fin}} \underbrace{A_{\text{fin}} h \theta_b}_{\dot{Q}_{\text{fin max}}}.$$

And the total heat transfer is

$$\begin{aligned} Q_{\text{total,fin}} &= \dot{Q}_{\text{fin}} + \dot{Q}_{\text{unfin}} \\ &= \eta_{\text{fin}} A_{\text{fin}} h \theta_b + A_{\text{unfin}} h \theta_b \\ &= (\eta_{\text{fin}} A_{\text{fin}} + A_{\text{unfin}})\, h \theta_b. \end{aligned}$$

The fin efficiency in the cases of *very long fins* and *adiabatic tips* can be expressed as:

$$\begin{aligned} \eta_{\text{ling fin}} &= \frac{1}{mL} \\ \eta_{\text{adiabatic tip}} &= \frac{\tanh mL}{mL}. \end{aligned}$$

3.6.4 Fin Effectiveness

There is no assurance that adding fins to a surface will *enhance* heat transfer. The performance of the fins is judged on the basis of the enhancement in heat transfer relative to the *no-fin* case. The performance of fins is expressed in terms of the fin effectiveness ϵ_{fin}:

$$\epsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} = \frac{\dot{Q}_{\text{fin}}}{h A_b \theta_b}.$$

Hence, A_b is the cross sectional area of the fin at the base and $\dot{Q}_{\text{no fin}}$ represents the rate of heat transfer from this area if no fin were attached. An effectiveness of $\epsilon_{\text{fin}} = 1$ indicates that the addiction of fins to the surface does not affect heat transfer at all.

The fin efficiency and fin effectiveness are different quantities but related to each other by:

$$\epsilon_{\text{fin}} = \frac{A_{\text{fin}}}{A_b} \theta_{\text{fin}}.$$

The overall effectiveness, $\epsilon_{\text{fin, overall}}$ is the effectiveness of the addition of fins to the entire area:

$$\epsilon_{\text{fin, overall}} = \frac{\dot{Q}_{\text{total, fin}}}{\dot{Q}_{\text{total, no fin}}} = \frac{\eta_{\text{fin}} A_{\text{fin}} + A_{\text{unfin}}}{A_b + A_{\text{unfin}}}.$$

3.7 Transient Heat Conduction

In transient heat conduction, the temperature field changes with time,

$$T = T(x, y, z, t)..$$

The governing heat equation therefore retains the transient term

$$\rho c \frac{\partial T}{\partial t}..$$

3.7.1 Lumped System Analysis

In heat transfer analysis, some bodies are observed to behave like a "lump", i.e. whose interior temperature remains uniform at any point during a heat transfer process. The temperature of such bodies can be taken to be a function of time only, $T(t)$. Analysis utilizing this idealization is known as *lumped system analysis*. This is often applicable for small bodies.

We consider a body of arbitrary shape of mass m, volume $\mathscr{V}$, surface area A_s, density ρ, and specific heat c_p initially at uniform temperature T_i. At time $t = 0$, the body is placed into a medium at temperature T_∞, and heat transfer takes place between the body and the medium with heat transfer coefficient h. We assume the system to be lumped, such that the temperature within the body remains uniform, $T = T(t)$. During a differential time element $\mathrm{d}t$, the temperature of the body rises by a differential amount $\mathrm{d}T$. An energy balance of the solid for the time interval $\mathrm{d}t$ can be expressed as

$$hA_s\,(T_\infty - T)\;\mathrm{d}t = mc_p\,\mathrm{d}T.$$

Noting that $m = \rho\mathscr{V}$ and $\mathrm{d}T = \mathrm{d}(T - T_\infty)$ since T_∞ = constant the above can be rearranged as

$$\frac{\mathrm{d}(T - T_\infty)}{T - T_\infty} = -\frac{hA_s}{\rho\mathscr{V}c_p}\,\mathrm{d}t.$$

integrating from $t = 0$, at which $T = T_i$ to a time t at which $T = T(t)$ gives

$$\ln\frac{T(t) - T_\infty}{T_i - T_\infty} = -\frac{hA_s}{\rho\mathscr{V}c_p}t.$$

Rearranging we obtain

$$T - T_\infty = (T_i - T_\infty)e^{-bt} \tag{10}$$

where

$$b = \frac{hA_s}{\rho\mathscr{V}c_p} \quad \mathrm{s}^{-1}$$

is a positive quantity. The reciprocal of b, is termed τ and is called the time constant defined as:

$$\tau = \frac{1}{b} = \frac{\rho c_p \mathscr{V}}{hA_s} \qquad \Longrightarrow \qquad \theta = e^{-\frac{t}{\tau}}.$$

Heat Transfer Rate and Total Heat Transfer Once the temperature $T(t)$ at time t is found from Equation (10), the rate of convection heat transfer between the body and its environment at

that time can be determined from Newton's law of cooling as

$$\dot{Q}(t) = hA_s\left(T(t) - T_\infty\right).$$

The *total amount* of heat transfer between the body and the surrounding medium over the time interval $t = 0$ to t is simply the change in the energy content of the body:

$$Q = mc_p(T(t) - T_i).$$

The amount of heat transfer reaches its upper limit when the body reaches the surrounding temperature, therefore the *maximum* heat transfer between the body and its surroundings is

$$Q_{\max} = mc_p(T_\infty - T_i).$$

3.7.2 Biot Number and Lumped-System Criterion

Now, we must determine when lumped analysis is appropriate to use. The first step in establishing such a criterion is to define a characteristic length as:

$$L_c = \frac{\mathscr{V}}{A_s}$$

and a dimensionless *Biot number* Bi as

$$\mathrm{Bi} = \frac{hL_c}{k}. \tag{11}$$

The characteristic length L_c to be used in the Biot number for simple geometries in which heat transfer is one-dimensional, such as a large plane wall of thickness $2L$, a long cylinder of radius r_o, and a sphere of radius r_o becomes L, $r_o/2$, and $r_o/3$, respectively. The Biot number is the ratio of internal resistance of a body to heat conduction to its external resistance to heat convection. Therefore a small Biot number represents a small resistance to conduction and thus small temperature gradients within the body.

Lumped system analysis assumes a uniform temperature distribution throughout the body. Thus lumped system analysis is exact when $\mathrm{Bi} = 0$ and approximate when $\mathrm{Bi} > 0$. It is generally accepted that lumped system analysis is applicable if

$$\mathrm{Bi} \leq 0.1.$$

3.7.3 One-Dimensional Transient Heat Conduction

We now move on to transient heat conduction in non-lumped bodies one-dimensional bodies. We now consider the variation of temperature with *time* and *position* in one-dimensional problems such as those associated with a large plane wall, a long cylinder, and a sphere.

Transient Conduction in a Plane Wall We consider a wall of total thickness $2L$. The x-coordinate is measured from the centre plane, so $x = 0$ is the middle and $x = L$ is the exposed face. The same fluid is on both sides.

The left side at $x = 0$ is insulated (i.e. no heat leaves) and the right side $x = L$ loses heat to the environment through convection. The initial state is everything starting at the same temperature.

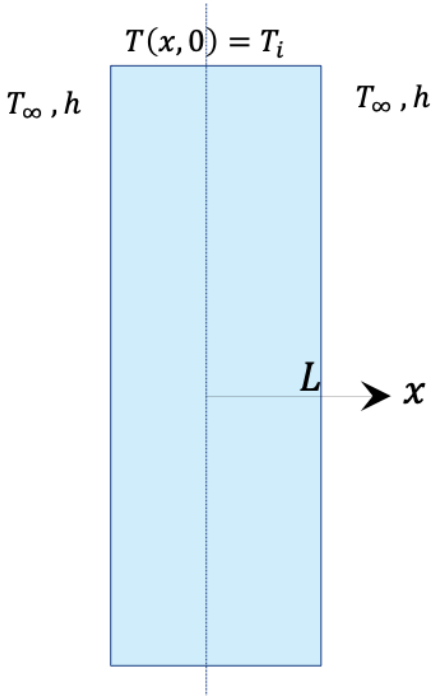


Figure 3.7: Transient Conduction in a Plane Wall.

Our governing differential equation is

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad \alpha = \frac{k}{\rho c_p}.$$

Dimensionless Temperature, Biot Number and Fourier Number Solving the above equation with the given BC's and IC would give T as a function of x, t, T_i, T_{inf}, k, α, h and L. Instead, we can also opt to use dimensional analysis to make the problem dimension-less. First, we define the dimensionless temperature:

$$\theta = \frac{T - T_\infty}{T_i - T_\infty}$$

this is the temperature excess.

We also define the Fourier number, Fo:

$$\text{Fo} = \frac{\alpha t}{L^2} = \frac{t}{L^2/\alpha}.$$

The Fourier number can be seen as the ratio of elapsed time to the time required for heat to diffuse through the body. So a small Fo means the process just started and a large Fo means diffusion has had a long time to act.

Using the Fourier and previously introduced Biot numbers the temperature excess of the wall can be written as:

$$\theta(x,t) = \frac{T - T_\infty}{T_i - T_\infty} = \underbrace{f_1(\text{FoBi})}_{\text{Time part}} \cdot \underbrace{f_2\left(\frac{x}{L}, \text{Bi}\right)}_{\text{Position part}}.$$

One-Term Approximation The exact solution to the above is an infinite series. For $\mathrm{Fo} > 0.2$ is the first term of the series, however, normally enough. Then:

$$f_1(\mathrm{Fo}, \mathrm{Bi}) \approx A_1 e^{-\lambda_1^2 \mathrm{Fo}}$$
$$f_2\left(\frac{x}{L}, \mathrm{Bi}\right) \approx \cos\left(\lambda_1 \frac{x}{L}\right).$$

Meaning

$$\theta = \frac{T - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \cdot \mathrm{Fo}} \cdot \cos\left(\lambda_1 \frac{x}{L}\right)$$

where, A_1 and λ_1 are functions of Bi which are tabulated in (Çengel and Ghajar 2015, Table 4–2). At the center of the wall, $x = 0$ the temperature T_0 is

$$\frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \mathrm{Fo}}.$$

When the center temperature has been determined, the spatial part can be used:

$$T(x,t) - T_\infty = (T_0(t) - T_\infty) \cos\left(\lambda_1 \frac{x}{L}\right).$$

So first, one must use the initial temperature to determine how much the center has heated/cooled at time t and then one uses the spatial profile to find the temperature near the surface or at any other point in the wall.

Instead of calculating the values of $A_1 e^{-\lambda_1^2 \mathrm{Fo}}$ and $\cos(\lambda_1 x/L)$ as shown above one can instead opt to read the values from so-called Heisler charts. These are just lookup-plots for the above equations.

Total Heat Transfer In lumped systems the entire body has uniform temperature so the total heat transfer is easily found, however know the temperature varies with position $T = T(x,t)$. In this case, the total energy change has to be integrated instead:

$$Q(t) = \int_{\mathscr{V}} \rho c (T(x,t) - T_i) \, \mathrm{d}\mathscr{V}.$$

After infinite time, the entire body reaches T_∞ and therefore:

$$Q_{\max} = mc(T_\infty - T_i)$$

so $Q/Q_{\max}$ tells the fraction of total energy transfer which has already happened. For the plane wall one-term solution:

$$\frac{Q}{Q_{\max}} = 1 - \theta_0 \frac{\sin \lambda_1}{\lambda_1}.$$

4 Fundamentals of Convection

In section 3: Conduction we have only considered conduction, i.e. the mechanism of heat transfer in a solid or still liquid. We now turn our focus to convection, which is the mechanism of heat transfer through a fluid in the presence of bulk fluid motion.

Convection can be classified as either *natural* or *free* and *forced convection* depending on how the motion of the fluid is initiated.

Definition 4.1: Natural or free convection

In natural convection, any fluid motion is caused by natural means such as buoyancy effects between a warmer and cooler fluid.

Definition 4.2: Forced convection

In forced convection, the fluid is forced to flow over a surface or through a pipe by external means such as a pump or a fan.

Convection may also be classified as either *external* or *internal*, depending on whether the fluid is forced to flow over a surface or through a pipe (Çengel and Ghajar 2015, p. 379).

4.1 The Energy Equation

Using energy conservation for an incompressible flow it can be shown that (Çengel and Ghajar 2015, Sec. 6–6)

$$\underbrace{\rho c \frac{\partial T}{\partial t}}_{\text{Unsteadiness}} + \underbrace{\rho c \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right)}_{\text{Advection}} = \underbrace{k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)}_{\text{Diffusion}} + \underbrace{\dot{e}_{\text{gen}}}_{\text{Heat Generation}} . \tag{12}$$

All the terms in (12) will be explained in section 4.1.1: Unsteadiness, section 4.1.2: Advection, section 4.1.3: Diffusion and section 4.1.4: Heat Generation.

4.1.1 Unsteadiness

The unsteadiness term $\rho c \partial T / \partial t$, describes how fast the temperature at a fixed point changes with time. Here, ρc is the energy stored per unit volume per degree of temperature rise. This is the only term in (12) that contains a time derivative and therefore the only term that becomes zero for steady-state.

4.1.2 Advection (Convection)

The advection or convection term $\rho c \boldsymbol{u} \cdot \nabla T$ is a measure of heat physically carried along by the moving fluid. For this term to survive, both a velocity $\boldsymbol{u}$ and a temperature gradient ∇T are needed. No motion means nothing carries the heat and uniform temperature means the moving fluid changes nothing. This term is the whole reason convection transfers heat far better than conduction.

4.1.3 Diffusion (Conduction)

The diffusion or conduction term $k \nabla^2 T$ describes the molecule-to-molecule conduction – it is exactly Fourier's law from (1). Positive curvature of the temperature means the element is cooler than its surroundings so heat flows in and the temperature rises.

4.1.4 Heat Generation

Heat generated inside the element is described by $\dot{e}_{\text{gen}}$. This covers resistive, chemical, nuclear, and all other forms of internal heating. It also contains viscous dissipation, i.e. friction between

fluid layers turning mechanical energy into heat.

4.2 Conservation of Mass and Momentum (Continuity and Navier-Stokes)

The continuity equation from fluid mechanics is a representation of conservation of mass. This can be represented as (Hansen 2026, Eq. (15))

$$\nabla \cdot \rho \boldsymbol{V} + \frac{\partial \rho}{\partial t} = 0. \tag{13}$$

For incompressible flow, which we are assuming in this analysis (13) reduces to

$$\nabla \cdot \boldsymbol{V} = 0. \tag{14}$$

Conservation of momentum is encapsulated by the Navier-Stokes equations, i.e. Newton's second law for a fluid element, represented as (Hansen 2026, Eqs. (27)–(28))

$$\rho \frac{\partial \boldsymbol{u}}{\partial t} + \rho (\boldsymbol{u} \cdot \nabla) \boldsymbol{u} = \mu \nabla^2 \boldsymbol{u} - \nabla p. \tag{15}$$

Here, the terms can be viewed in the same manner as for (12) with an unsteadiness, advection, viscous and pressure term representing the same basic ideas as the corresponding terms in (12).

To use (12) one must first apply the four equations represented by (14) and (15). This gives the three velocity composants u, v, w and the pressure p. Only then can the energy equation (12) be used to solve for T.

4.3 Reynolds Number

As explained in the fluid mechanics course (Hansen 2026) flow can either be turbulent or laminar and normally slow velocity flows will stay laminar whereas faster flows quickly turn chaotic. This is described by the Reynolds number, Re, defined as the ratio of inertia forces to viscous forces as (Çengel and Ghajar 2015, Eq. (6–13))

$$\mathrm{Re} = \frac{V L_c}{\nu} = \frac{\rho V L_c}{\mu}$$

where V is the upstream velocity, L_c the characteristic length, and $\nu = \mu/\rho$ is the kinematic viscosity of the plate. Likewise, the local Reynolds number Re_x is defined as

$$\mathrm{Re}_x = \frac{V x}{\nu},$$

where x is the distance from the plates leading edge.

The Reynolds number at which a flow turns turbulent is called the *critical Reynolds number*, the size of which is determined by the geometry and flow conditions. For flow over a flat plate the generally accepted value for the critical Reynolds number is $\mathrm{Re}_{\mathrm{cr}} = V x_{\mathrm{cr}}/\nu = 5 \cdot 10^5$, where x_{cr} is the distance from the leading edge of the plate at which transition from laminar to turbulent flow occurs (Çengel and Ghajar 2015, pp. 391–392).

4.4 Prandtl Number

The relative thickness of the velocity and thermal boundary layers is best described by the dimensionless parameter Prandtl number, Pr, defined as (Çengel and Ghajar 2015, Eq. (6–12)).

$$\mathrm{Pr} = \frac{\nu}{\alpha} = \frac{\mu c_p}{k}.$$

The Prandtl numbers of fluids range from less than 0.01 for liquid metals to more than 100 000 for heavy oils. The Prandtl number for water is Pr = 1.7–13.7 (Çengel and Ghajar 2015, Tab. 6–2).

The Prandtl numbers of gasses are about 1, which indicate that moth momentum and heat dissipate through the fluid at the same rate. Heat diffuses very quickly in liquid metal ($\mathrm{Pr} \ll 1$) and very slowly in oils ($\mathrm{Pr} \gg 1$) relative to momentum. Hence, the thermal boundary layer is much thicker in a liquid metal and much thinner in oils relative to the velocity boundary layer (Çengel and Ghajar 2015, p. 390).

4.5 Nusselt Number

In convection, the heat transfer coefficient h is often nondimensionalized through the Nusselt number, defined as (Çengel and Ghajar 2015, Eq. (6–5)):

$$\mathrm{Nu} = \frac{hL_c}{k}.$$

The Nusselt number can be seen as the ratio between the convective heat flux $\dot{q}_{\mathrm{conv}}$ and the conductive heat flux $\dot{q}_{\mathrm{cond}}$ as (Çengel and Ghajar 2015, Eqs. (6–6—6–8))

$$\begin{aligned}
\dot{q}_{\mathrm{conv}} &= h\Delta t \\
\dot{q}_{\mathrm{cond}} &= k\frac{\Delta T}{L} \\
\frac{\dot{q}_{\mathrm{conv}}}{\dot{q}_{\mathrm{cond}}} &= \frac{hL}{k} = \mathrm{Nu},
\end{aligned}$$

hence, the Nusselt number represents the enhancement of heat transfer through a fluid layer as a result of convection relative to static conduction across the same fluid layer. A large Nusselt number implies a more effective convection and a Nusselt number of Nu = 1 represents purely conductive heat transfer across a fluid layer.

4.6 Boundary Layers in Heat Transfer

In Fluid mechanics it was shown how a velocity boundary layer arises when a fluid flow over a surface as a result of the fluid velocity being zero at the interface (Hansen 2026). Likewise, for a flow of a fluid at a uniform temperature T_∞ over an isothermal plate at temperature T_s, the temperature profile in the flow field ranges from T_s at the surface to T_∞ sufficiently far from the interface. Normally the thickness of the thermal boundary layer δ_t at any location along the surface is defined as the distance from the surface at which the temperature difference $T - T_s = 0.99(T_\infty - T_s)$, i.e. 99 % of the temperature difference is already "covered". The rate of convective heat transfer is related to the temperature gradient at any given location along the surface, hence the shape of the temperature profile in the thermal boundary layer dictates the convection heat transfer between the surface and the fluid (Çengel and Ghajar 2015, p. 389).

The thickness of the velocity boundary layer δ over a flat plate, defined as the distance from the surface for which $u/V = 0.99$ is given by (Çengel and Ghajar 2015, Eq. (6–51))

$$\delta = \frac{4.91}{\sqrt{\frac{V}{\nu x}}} = \frac{4.91x}{\sqrt{\mathrm{Re}_x}}$$

where x is the distance from the leading edge of the plate. It is worth noting that the boundary layer thickness increases with kinematic viscosity ν and with increasing distance from the leading edge x, but decreases with increasing free-stream velocity V (Çengel and Ghajar 2015, p. 403). Conversely, the thickness of the thermal boundary layer δ_t, for laminar flow over an isothermal flat plate is given by (Çengel and Ghajar 2015, Eq. (6–63))

$$\delta_t = \frac{\delta}{\sqrt[3]{\mathrm{Pr}}} = \frac{4.91x}{\sqrt[3]{\mathrm{Pr}}\sqrt{\mathrm{Re}_x}}$$

Where Pr is the Prandtl number.

4.6.1 Local Skin Friction and Heat Transfer

For laminar flow over an isothermal flat plate, the local skin-friction coefficient is

$$C_{f,x} = \frac{\tau_w}{\frac{1}{2}\rho V^2} = \frac{0.664}{\sqrt{\mathrm{Re}_x}}.$$

Here, τ_w is the wall shear stress and V is the free-stream velocity.

For the same flow, the local Nusselt number is

$$\mathrm{Nu}_x = \frac{h_x x}{k} = 0.332\,\mathrm{Re}_x^{1/2}\mathrm{Pr}^{1/3}, \qquad \mathrm{Pr} > 0.6.$$

Here, h_x is the local convection heat-transfer coefficient.

4.6.2 Stanton Number and Reynolds Analogies

The Stanton number relates the local convection heat-transfer coefficient to the heat-capacity flow of the fluid:

$$\mathrm{St}_x = \frac{h_x}{\rho c_p V} = \frac{\mathrm{Nu}_x}{\mathrm{Re}_x\mathrm{Pr}}.$$

For flow with $\mathrm{Pr} = 1$, the Reynolds analogy relates heat transfer to skin friction:

$$\mathrm{St}_x = \frac{C_{f,x}}{2}.$$

The Chilton–Colburn analogy extends this comparison to other Prandtl numbers:

$$j_H = \mathrm{St}_x\mathrm{Pr}^{2/3} = \frac{C_{f,x}}{2}.$$

Equivalently,

$$\frac{C_{f,x}}{2}\,\mathrm{Re}_x = \mathrm{Nu}_x\mathrm{Pr}^{-1/3}.$$

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5 External Forced Convection

5.1 Parallel Flow Over Flat Plates

We consider the parallel flow of a fluid over a flat plate of length L in the flow direction as shown in figure 5.1. Here, the x-coordinate is measured along the plate surface from the leading edge in the flow-direction. The fluid approaches the plate in the x-direction with a uniform velocity V and temperature T_∞. The flow in the boundary layer is initially laminar, but becomes turbulent at a distance x_{cr} from the leading edge where the Reynolds number reaches its critical value for transition – For flow over a flat plate transition occurs at about $\mathrm{Re} \approx 5 \cdot 10^5$ (Çengel and Ghajar 2015, pp. 428–429).

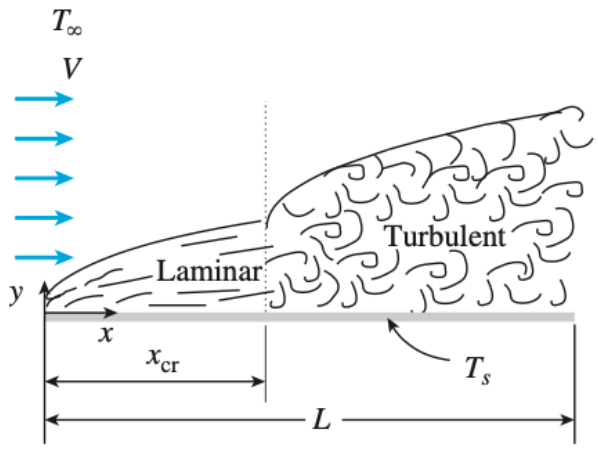


Figure 5.1: Laminar and turbulent flowregions of the boundary layer during flow over a flat plate. Reproduced from (Çengel and Ghajar 2015, Fig. 7–6).

5.1.1 Friction Coefficient

For laminar flow over a plate the boundary layer coefficient is analytically determined to be (Çengel and Ghajar 2015, Eq. (7–12a, b)):

$$\delta = \frac{4.91x}{\sqrt{\mathrm{Re}_x}}, \quad C_{f,x} = \frac{0.664}{\sqrt{\mathrm{Re}_x}}, \quad \mathrm{Re}_x < 5 \cdot 10^5. \tag{16}$$

From experiments the corresponding relations for turbulent flow are found to be (Çengel and Ghajar 2015, Eq. (7–13a, b))

$$\delta = \frac{0.38x}{\sqrt[5]{\mathrm{Re}_x}}, \quad C_{f,x} = \frac{0.059}{\sqrt[5]{\mathrm{Re}_x}}, \quad 5 \cdot 10^5 \leq \mathrm{Re}_x \leq 10^7. \tag{17}$$

Note the local friction coefficient $C_{f,x} \propto \mathrm{Re}_x^{-1/2}$ and thus to $x^{-1/2}$ for laminar flow. Hence, $C_{f,x}$ is supposedly infinite at the leading edge and decreases by a factor of $x^{-1/2}$ in the flow direction. Also, note that $C_{f,x}$ reaches its highest value when the flow turns fully turbulent, where after it decreases by a factor of $x^{-1/5}$ in the flow direction.

The average friction coefficient over the entire surface is then found using integration over the entire length (Çengel and Ghajar 2015, Eq. (7–7))

$$C_f = \frac{1}{L}\int_0^L C_{f,x}\,\mathrm{d}x. \tag{18}$$

Substituting (16) and (17) into (18) and performing the integration we obtain (Çengel and Ghajar 2015, Eqs. (7–14)–(7–15))

$$\text{Laminar:} \qquad C_f = \frac{1.33}{\sqrt{\mathrm{Re}_L}}, \qquad \mathrm{Re}_L < 5\cdot 10^5 \tag{19}$$

$$\text{Turbulent:} \qquad C_f = \frac{0.074}{\sqrt[5]{\mathrm{Re}_L}}, \qquad 5\cdot 10^5 \leq \mathrm{Re}_L \leq 10^7. \tag{20}$$

Equation (19) gives the average friction coefficient for the entire plate when the flow is laminar over the entire plate, note that this is twice the value of the local friction coefficient at the end of the plate, $C_f = 2C_{f,x=L}$. Equation (20) gives the average friction over the entire plate when the flow is turbulent over the entire plate or when the laminar flow region is small enough to be neglected $x_{\mathrm{cr}} \ll L$.

To obtain the actual friction coefficient over a plate with both a laminar and turbulent region, the integration in (18) is carried out over the laminar region using (16) and the turbulent region using (17) as (Çengel and Ghajar 2015, Eq. (7–16))

$$C_f = \frac{1}{L}\left(\int_0^{x_{\mathrm{cr}}} C_{f,x,\mathrm{laminar}}\,\mathrm{d}x + \int_{x_{\mathrm{cr}}}^{L} C_{f,x,\mathrm{turbulent}}\,\mathrm{d}x\right). \tag{21}$$

Taking the critical Reynolds number to be $\mathrm{Re}_{\mathrm{cr}} = 5\cdot 10^5$ the integration in (21) is solved to obtain the following expression for the average friction coefficient over the entire plate

$$C_f = \frac{0.074}{\sqrt[5]{\mathrm{Re}_L}} - \frac{1742}{\mathrm{Re}_L}, \quad 5\cdot 10^5 \leq \mathrm{Re}_L \leq 10^7. \tag{22}$$

In (22) the surfaces are assumed to be smooth and the free stream is assumed turbulence tree.

For laminar flow, the friction coefficient only depends on Reynolds number, whereas for turbulent flow, surface roughness causes the friction coefficient to increase many times. Experimental observations give a average friction coefficient for turbulent flow over rough surfaces as:

$$C_f = \left(1.89 - 1.62\log\frac{\epsilon}{L}\right)^{-2.5},$$

where ϵ is the surface roughness and L is the length of the plate in the flow direction. This relation is applicable for turbulent flow on rough surfaces for $\mathrm{Re} > 10^6$ especially when $\epsilon/L > 10^{-4}$ (Çengel and Ghajar 2015, pp. 429–430).

5.1.2 Heat Transfer Coefficient

The locals Nusselt number at a location x for laminar flow over a flat plate can be found analytically by solving the differential energy equation to be (Çengel and Ghajar 2015, Eq. (7–19))

$$\mathrm{Nu}_x = \frac{h_x x}{k} = 0.332\sqrt{\mathrm{Re}_x}\sqrt[3]{\mathrm{Pr}}, \quad \mathrm{Pr} > 0.6, \mathrm{Re}_x < 5\cdot 10^5..$$

From experiments, the corresponding relation for turbulent flow is (Çengel and Ghajar 2015, Eq. (7–20))

$$\mathrm{Nu}_x = \frac{h_x x}{k} = 0.0296 \mathrm{Re}_x^{0.8} \sqrt[3]{\mathrm{Pr}}, \quad 0.6 \leq \mathrm{Pr} \leq 60, 5 \cdot 10^5 \leq \mathrm{Re}_x \leq 10^7.$$

Like, for the friction coefficient h_x is, for laminar flow, proportional to $\sqrt{\mathrm{Re}_x}$ and thus to $x^{-1/2}$, meaning h_x is supposedly infinite at the leading edge and decreases by a factor of $x^{-1/2}$ in the flow direction.

Like for the friction coefficient the average convection coefficient over the entire surface is found through integration (Çengel and Ghajar 2015, Eq. (7–8))

$$h = \frac{1}{L} \int_0^L h_x \, \mathrm{d}x.$$

Hence, the average Nusselt number over the entire plate is (Çengel and Ghajar 2015, Eqs. (7–21)–(7–22))

$$\text{Laminar:} \qquad \mathrm{Nu} = \frac{hL}{k} = 0.664 \sqrt{\mathrm{Re}_L} \sqrt[3]{\mathrm{Pr}}, \qquad \mathrm{Re}_L < 5 \cdot 10^5, \mathrm{Pr} > 0.6 \tag{23}$$

$$\text{Turbulent:} \qquad \mathrm{Nu} = \frac{hL}{k} = 0.037 \mathrm{Re}_L^{0.8} \sqrt[3]{\mathrm{Pr}}, \qquad 5 \cdot 10^5 \leq \mathrm{Re}_L \leq 10^7, 0.6 \leq \mathrm{Pr} \leq 60. \tag{24}$$

Here, the same conditions apply as for the friction coefficient in that the laminar result only applies for fully laminar flow and vice versa for the turbulent result. An accurate heat transfer coefficient can be found by performing the same integration as for the friction coefficient. Taking the critical Reynolds number to be $\mathrm{Re}_{\mathrm{cr}} = 5 \cdot 10^5$ the resulting average Nusselt number over the entire plate is (Çengel and Ghajar 2015, Eq. (7–24))

$$\mathrm{Nu} = \frac{hL}{k} = \left(0.037 \mathrm{Re}_L^{0.8} - 871\right) \sqrt[3]{\mathrm{Pr}}, \quad 0.6 \leq \mathrm{Pr} \leq 60, 5 \cdot 10^5 \leq \mathrm{Re}_L \leq 10^7.$$

Using curve-fitting the following relation for the Nusselt number, which is claimed to be applicable for all Prandtl numbers with an accuracy of ±1 % (Çengel and Ghajar 2015, Eq. (7–26), pp. 430–432)

$$\mathrm{Nu}_x = \frac{h_c x}{k} = \frac{0.3387 \sqrt[3]{\mathrm{Pr}} \sqrt{\mathrm{Re}_x}}{\sqrt[4]{1 + (0.0468/\mathrm{Pr})^{2/3}}}, \qquad \mathrm{Re}_x \mathrm{Pr} \geq 100.$$

5.2 Flow Across Cylinders and Spheres

The characteristic length for a circular cylinder or sphere is the external diameter D. Thus, the Reynolds number for a circular cylinder or sphere is $\mathrm{Re} = \mathrm{Re}_D = VD/\nu$, where V is the velocity as it approaches the cylinder or sphere. The critical Reynolds number for flow across a circular cylinder or sphere is about $\mathrm{Re}_{\mathrm{cr}} \approx 2 \cdot 10^5$ (Çengel and Ghajar 2015, p. 438).

5.2.1 Heat Transfer Coefficient

Flows across cylinders and spheres generally involve flow separation and are thus difficult to analyze analytically. Hence, such flows much be studied experimentally or numerically. The average Nusselt number for cross flow over a cylinder is given by (Çengel and Ghajar 2015,

Eq. (7–35))

$$\mathrm{Nu}_{\mathrm{cyl}} = \frac{hD}{k} = 0.3 + \frac{0.62\sqrt{\mathrm{Re}}\sqrt[3]{\mathrm{Pr}}}{\sqrt[4]{1 + (0.4/\mathrm{Pr})^{2/3}}}\left(1 + \left(\frac{\mathrm{Re}}{282\,000}\right)^{5/8}\right)^{4/5}.$$

For flow over a sphere, the average Nusselt number is (Çengel and Ghajar 2015, Eq. (7–36))

$$\mathrm{Nu}_{\mathrm{sph}} = \frac{hD}{k} = 2 + \left(0.4\sqrt{\mathrm{Re}} + 0.06\mathrm{Re}^{2/3}\right)\mathrm{Pr}^{0.4}\sqrt[4]{\frac{\mu_\infty}{\mu_s}},$$
$$3.5 \le \mathrm{Re} \le 8 \cdot 10^4, \quad 0.7 \le \mathrm{Pr} \le 380, \quad 1.0 \le \mu_\infty/\mu_s \le 3.2.$$

All the fluid properties in this case are evaluated at the free-stream temperature T_∞, except for μ_s, which is evaluated at the surface temperature T_s.

The average Nusselt number for flow across cylinders can be expressed compactly as (Çengel and Ghajar 2015, Eq. (7–37))

$$\mathrm{Nu}_{\mathrm{cyl}} = \frac{hD}{k} = C\mathrm{Re}^m\mathrm{Pr}^n,$$

where $n = 1/3$ and the experimentally determined constants C and m are given in table 5.1 (Çengel and Ghajar 2015, pp. 442–443).

Table 5.1: Empirical correlations for the average Nusselt number for forced convection over circular and noncircular cylinders in crossflow. Adapted from (Çengel and Ghajar 2015, Tab. 7–1).

Cross-section of cylinder	Fluid	Range of Re	Nusselt number
Circle	Gas or liquid	0.4–4	$\mathrm{Nu} = 0.989\mathrm{Re}^{0.330}\mathrm{Pr}^{1/3}$
		4–40	$\mathrm{Nu} = 0.911\mathrm{Re}^{0.385}\mathrm{Pr}^{1/3}$
		40–4000	$\mathrm{Nu} = 0.683\mathrm{Re}^{0.466}\mathrm{Pr}^{1/3}$
		4000–40 000	$\mathrm{Nu} = 0.193\mathrm{Re}^{0.618}\mathrm{Pr}^{1/3}$
		40 000–400 000	$\mathrm{Nu} = 0.027\mathrm{Re}^{0.805}\mathrm{Pr}^{1/3}$
Square	Gas	3900–79 000	$\mathrm{Nu} = 0.094\mathrm{Re}^{0.675}\mathrm{Pr}^{1/3}$
Square, tilted 45°	Gas	5600–111 000	$\mathrm{Nu} = 0.258\mathrm{Re}^{0.588}\mathrm{Pr}^{1/3}$
Hexagon	Gas	4500–90 700	$\mathrm{Nu} = 0.148\mathrm{Re}^{0.638}\mathrm{Pr}^{1/3}$
Hexagon, tilted 45°	Gas	5200–20 400	$\mathrm{Nu} = 0.162\mathrm{Re}^{0.638}\mathrm{Pr}^{1/3}$
		20 400–105 000	$\mathrm{Nu} = 0.039\mathrm{Re}^{0.782}\mathrm{Pr}^{1/3}$
Vertical plate	Gas	6300–23 600	$\mathrm{Nu} = 0.257\mathrm{Re}^{0.731}\mathrm{Pr}^{1/3}$
Ellipse	Gas	1400–8200	$\mathrm{Nu} = 0.197\mathrm{Re}^{0.612}\mathrm{Pr}^{1/3}$

5.2.2 Flow Across Tube Banks

Flow through tube banks is studied experimentally as it is too complex to be treated analytically. The average Nusselt number over the entire tube bank is (Çengel and Ghajar 2015, Eq. (7–42))

$$\mathrm{Nu}_D = \frac{hD}{k}C\mathrm{Re}_D^m\mathrm{Pr}^n\sqrt[4]{\mathrm{Pr}/\mathrm{Pr}_s},$$

where D is the tube diameter and the values of the constants C, m, and n depend on the Reynolds number. Such correlations are given in table 5.2 for tube banks with more than 16 rows $N_L > 16$, $0.7 < \mathrm{Pr} < 500$ and $0 < \mathrm{Re}_D < 2 \cdot 10^6$ with an uncertainty of ±15 %. Note that all properties except

Pr_s are evaluated at the arithmetic mean temperature T_m of the fluid determined from (Çengel and Ghajar 2015, Eq. (7–43))

$$T_m = \frac{T_i + T_e}{2}$$

where T_i and T_e are the fluid inlet and exit temperatures of the tube bank respectively.

The average Nusselt number relations given in table 5.2 are only applicable for tube banks with more than 16 rows. These relations can also be used for tube banks with $N_L < 16$ provided they are modified using (Çengel and Ghajar 2015, Eq. (7–44))

$$\mathrm{Nu}_{D,N_{L<16}} = F\mathrm{Nu}_D, \tag{25}$$

where F is a correction factor whose values are given in table 5.3. For $\mathrm{Re}_D > 1000$ the correction factor is independent of Reynolds number.

Table 5.2: Nusselt number correlations for crossflow over tube banks for $N_L > 16$ and $0.7 < \mathrm{Pr} < 500$. Adapted from (Çengel and Ghajar 2015, Tab. 7–2).

Arrangement	Range of Re_D	Correlation
In-line	0–100	$\mathrm{Nu}_D = 0.9\mathrm{Re}_D^{0.4}\mathrm{Pr}^{0.36}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$
	100–1000	$\mathrm{Nu}_D = 0.52\mathrm{Re}_D^{0.5}\mathrm{Pr}^{0.36}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$
	$1000–2 \cdot 10^5$	$\mathrm{Nu}_D = 0.27\mathrm{Re}_D^{0.63}\mathrm{Pr}^{0.36}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$
	$2 \cdot 10^5–2 \cdot 10^6$	$\mathrm{Nu}_D = 0.033\mathrm{Re}_D^{0.8}\mathrm{Pr}^{0.4}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$
Staggered	0–500	$\mathrm{Nu}_D = 1.04\mathrm{Re}_D^{0.4}\mathrm{Pr}^{0.36}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$
	500–1000	$\mathrm{Nu}_D = 0.71\mathrm{Re}_D^{0.5}\mathrm{Pr}^{0.36}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$
	$1000–2 \cdot 10^5$	$\mathrm{Nu}_D = 0.35(S_T/S_L)^{0.2}\mathrm{Re}_D^{0.6}\mathrm{Pr}^{0.36}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$
	$2 \cdot 10^5–2 \cdot 10^6$	$\mathrm{Nu}_D = 0.031(S_T/S_L)^{0.2}\mathrm{Re}_D^{0.8}\mathrm{Pr}^{0.36}(\mathrm{Pr}/\mathrm{Pr}_s)^{0.25}$

Table 5.3: Correction factor F for use in (25) for $N_L < 16$ and $\mathrm{Re}_D > 1000$. Adapted from (Çengel and Ghajar 2015, Tab. 7–3).

N_L	1	2	3	4	5	7	10	13
In-line	0.70	0.80	0.86	0.90	0.93	0.96	0.98	0.99
Staggered	0.64	0.76	0.84	0.89	0.93	0.96	0.98	0.99

Once the Nusselt number and therefore the heat transfer coefficient for the entire tube bank is known, the heat transfer rate can be determined from Newton's law of cooling using a suitable temperature difference ΔT. This suitable temperature difference is, for internal flow, the log mean temperature difference ΔT_{1m} defined as (Çengel and Ghajar 2015, Eq. (7–45))

$$\Delta T_{1m} = \frac{(T_s - T_e) - (T_s - T_i)}{\ln\left(\frac{T_s - T_e}{T_s - T_i}\right)} = \frac{\Delta T_e - \Delta T_i}{\ln\left(\frac{\Delta T_e}{\Delta T_i}\right)}.$$

Also, the exit temperature of the fluid T_e can be determined from (Çengel and Ghajar 2015, Eq. (7–46))

$$T_e = T_s - (T_s - T_i)\exp\left(-\frac{A_s h}{\dot{m} c_p}\right),$$

where $A_s = N\pi DL$ is the heat transfer surface area and $\dot{m} = \rho V N_t S_T L$ is the mass flow rate of the fluid. Here, N is the total number of tubes in the bank, i.e. the product of N_T and N_L which are the number of tubes in the transverse plane and longitudinal plane respectively, L is the length of

the tubes, and V is the velocity of the fluid just before entering the tube bank. The heat transfer rate can be determined from (Çengel and Ghajar 2015, Eq. (7–47))

$$\dot{Q} = hA_s\Delta T_{1m} = \dot{m}c_p(T_e - T_i).$$

In the above the second relation is usually more convenient as it does not require the calculation of ΔT_{1m} (Çengel and Ghajar 2015, pp. 446–449).

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6 Internal Forced Convection

In lecture 6 we covered external flows, where the fluid has a free surface and thus the boundary layer is free to grow indefinitely. In this lecture we will consider internal flow, where the fluid is completely confined by the inner surfaces of a tube, which limits the growth of the boundary layer. For a more general description of internal flows than the one provided in the following sections, I refer to my Fluid Mechanics notes (Hansen 2026, Ch. 18).

6.1 Average Velocity and Temperature

In internal flow, there is no free stream and thus no free-stream velocity. An often used alternative for this is the *average velocity* V_{avg}, which, for incompressible flow, remains constant for constant tubular cross-section. The average velocity can be found using the principle of conservation of mass (Çengel and Ghajar 2015, Eq. (8–1))

$$\dot{m} = \rho V_{\text{avg}} A_c = \int_{A_c} \rho u(r)\,\mathrm{d}A_c$$

and for incompressible flow in a circular pipe of radius R, the aberage velocity is (Çengel and Ghajar 2015, Eq. (8–2))

$$V_{\text{avg}} = \frac{2}{R^2}\int_0^R u(r) r\,\mathrm{d}r.$$

Hence, the average velocity can either be found using the flow rate or the velocity profile.

Likewise, a fluid moving through a tube undergoing heating or cooling will have different temperatures across the cross section of the tube. Therefore, it is convenient to introduce the *average temperature* T_{avg}, which remains constant at a cross section. Opposed to the average velocity, the average temperature changes in the flow direction when the fluid is heated or cooled. The value of the average temperature can be determined using the principle of conservation of energy as (Çengel and Ghajar 2015, Eq. (8–3))

$$\dot{E}_{\text{fluid}} = \dot{m}c_pT_{\text{avg}} = \int_{\dot{m}} c_pT(r)\delta\dot{m} = \int_{A_c} \rho c_pT(r)u(r)\,\mathrm{d}A_c.$$

For a fluid with constant density and specific heat flowing in a circular pipe of radius R, the average temperature can be expressed as (Çengel and Ghajar 2015, Eq. (8–4))

$$T_{\text{avg}} = \frac{2}{V_{\text{avg}}R^2}\int_0^R T(r)u(r)r\,\mathrm{d}r.$$

The fluid properties in internal flow is conveniently evaluated at the *bulk mean fluid temperature* T_b which is the arithmetic mean of the average temperatures at the inlet T_i and the exit T_e (Çengel and Ghajar 2015, pp. 475–476)

$$T_b = \frac{T_i + T_e}{2}.$$

6.1.1 Laminar and Turbulent Flow in Tubes

Flow in a tube may, like external flow, be either laminar or turbulent depending on the flow conditions. Most pipe flows are in practice turbulent and laminar flow is encountered only when highly viscous fluids such as oils flow in small diameter tubes. For flow in a circular tube, the Reynolds number is given by (Çengel and Ghajar 2015, Eq. (8–5))

$$\mathrm{Re} = \frac{V_{\mathrm{avg}} D}{\nu} = \frac{\rho V_{\mathrm{avg}} D}{\mu} = \frac{\rho D}{\mu}\left(\frac{\dot{m}}{\rho \pi \frac{D^2}{4}}\right) = \frac{4\dot{m}}{\mu \pi D},$$

where D is the tube diameter.

For flow through non-circular tubes, the Reynolds number, the Nusselt number, and the friction factor are based on the *hydraulic diameter* D_h defined as (Çengel and Ghajar 2015, Eq. (8–6))

$$D_h = \frac{4A_c}{p},$$

where A_c is the cross-sectional area of the tube and p is its wetted perimeter.

Under most normal conditions, flow in tubes is laminar for $\mathrm{Re} < 2300$ and fully turbulent for $10\,000 < \mathrm{Re}$ (Çengel and Ghajar 2015, p. 476).

Laminar Flow For Laminar flow in tubes of various geometries the friction factors and Nusselt numbers are given in table 6.1 (Çengel and Ghajar 2015, p. 491)

Turbulent Flow It is often desirable to use turbulent flow due to the higher heat transfer coefficients asssociated with it.

For smooth tubes, the friction factor for turbulent flow can be determined from the first Pethukov equation (Çengel and Ghajar 2015, Eq. (8–65))

$$f_{\text{smooth tube}} = (0.790 \ln \mathrm{Re} - 1.64)^{-2}, \qquad 3000 < \mathrm{Re} < 5 \cdot 10^6. \tag{26}$$

The Nusselt number in turbulent flow is related to the friction factor through the Chilton-Colburn analogy as (Çengel and Ghajar 2015, Eq. (8–66))

$$\mathrm{Nu} = 0.125 f \mathrm{Re} \sqrt[3]{\mathrm{Pr}}. \tag{27}$$

Equation (27) is applicable for both smooth and rough tubes. For fully developed turbulent flow in smooth tubes, the Nusselt number can be found using the Dittus-Boelter equation (Çengel and Ghajar 2015, Eq. (8–68))

$$\mathrm{Nu}_{\text{smooth}} = 0.023 \mathrm{Re}^{0.8} \mathrm{Pr}^n \tag{28}$$

where $n = 0.4$ for heating and $n = 0.3$ for cooling.

Table 6.1: Nusselt number and friction factor for fully developed laminar flow in tubes of various cross-sections. Adapted from (Çengel and Ghajar 2015, Tab. 8–1).

Tube geometry	a/b or θ	*Nusselt number*		Friction factor f
		T_s = const.	$\dot{q}_s$ = const.	
Circle	—	3.66	4.36	64.00/Re
Rectangle	1	2.98	3.61	56.92/Re
	2	3.39	4.12	62.20/Re
	3	3.96	4.79	68.36/Re
	4	4.44	5.33	72.92/Re
	6	5.14	6.05	78.80/Re
	8	5.60	6.49	82.32/Re
	∞	7.54	8.24	96.00/Re
Ellipse	1	3.66	4.36	64.00/Re
	2	3.74	4.56	67.28/Re
	4	3.79	4.88	72.96/Re
	8	3.72	5.09	76.60/Re
	16	3.65	5.18	78.16/Re
Isosceles triangle	10°	1.61	2.45	50.80/Re
	30°	2.26	2.91	52.28/Re
	60°	2.47	3.11	53.32/Re
	90°	2.34	2.98	52.60/Re
	120°	2.00	2.68	50.96/Re

A more accurate relation than (28) is the Gnielinski equation (Çengel and Ghajar 2015, Eq. (8–71))

$$\mathrm{Nu} = \frac{\frac{f}{8}(\mathrm{Re} - 1000)\,\mathrm{Pr}}{1 + 12.7\sqrt{\frac{f}{8}}\left(\sqrt[3]{\mathrm{Pr}^2} - 1\right)} \qquad 0.5 \leq \mathrm{Pr} \leq 2000 \quad 3 \cdot 10^3 < \mathrm{Re} < 5 \cdot 10^6,$$

where the friction factor must first be determined from another relation such as the first Petukhov equation (26).

The above relations do not apply to liquid metals due to their very low Prandtl numbers. For liquid metals with $0.004 < \mathrm{Pr} < 0.01$ the Sleicher-Rouse equations are recommended for $10^4 < \mathrm{Re} < 10^6$ (Çengel and Ghajar 2015, Eqs. (8–72) & (8–73))

$$T_s = \text{constant}: \qquad \mathrm{Nu} = 4.8 + 0.0156\mathrm{Re}^{0.85}\mathrm{Pr}_s^{0.93},$$
$$\dot{q}_s = \text{constant}: \qquad \mathrm{Nu} = 6.3 + 0.0167\mathrm{Re}^{0.85}\mathrm{Pr}_s^{0.93},$$

where the subscript s indicates that the Prandtl number is to be evaluated at the surface temperature (Çengel and Ghajar 2015, pp. 496–497).

6.2 The Entrance Region and Entry Lengths

As explained in (Hansen 2026, Ch. 17) flow in a pipe will at some point become fully-developed at which point the velocity profile stops changing along the length of the pipe. The region between the inlet and the point at which the velocity profile is fully-developed is termed the *hydrodynamic entrance region* and the length of this region is the *hydrodynamic entry length* L_h

The region of flow over which the thermal boundary layer develops and reaches the tube center is

termed the *thermal entrance region* and the length of this region is called the *thermal entry length* L_t.

In laminar flow the hydrodynamic and thermal entry lengths are approximately (Çengel and Ghajar 2015, Eqs. (8–11) & (8–12))

$$L_{h,\text{laminar}} \approx 0.05\text{Re}D$$
$$L_{t,\text{laminar}} \approx 0.05\text{RePr}D = \text{Pr}L_{h,\text{laminar}}.$$

In turbulent flow the hydrodynamic and thermal entry lengths are approximately of the same size due to the intense mixing that occurs and it relies a lot less on the Reynolds number instead being given as (Çengel and Ghajar 2015, Eq. (8–13), pp. 477–479)

$$L_{h,\text{turbulent}} \approx L_{t,\text{turbulent}} \approx 10D.$$

6.3 Thermal Analysis

In the absence of work interactions, the conservation of energy equation for steady flow of a fluid in a tube can be expressed as (Çengel and Ghajar 2015, Eq. (8–14))

$$\dot{Q} = \dot{m}c_p\left(T_e - T_i\right),$$

where T_i and T_e are the average fluid temperatures at the inlet and exit of the tube, respectively, and $\dot{Q}$ is the rate of heat transfer to or from the fluid.

The surface heat flux can be expressed as (Çengel and Ghajar 2015, Eq. (8–15))

$$\dot{q}_s = h_x\left(T_s - T_{\text{avg}}\right),$$

where h_x is the *local* heat transfer coefficient and T_s and T_{avg} are the surface and average fluid temperatures at that location respectively (Çengel and Ghajar 2015, pp. 480–481).

6.3.1 Constant Surface Heat Flux

For constant surface heat flux $\dot{q}_s$ = constant like in the case of uniform resistive heating the rate of heat transfer is (Çengel and Ghajar 2015, Eq. (8–16))

$$\dot{Q} = \dot{q}_s A_s = \dot{m}c_p(T_e - T_i)$$

and the average fluid temperature at the tube exit is (Çengel and Ghajar 2015, Eq. (8–17))

$$T_e = T_i + \frac{\dot{q}_s A_s}{\dot{m}c_p}.$$

Hence, the average fluid temperature will increase linearly in the flow direction in the case of constant surface heat flux, since the surface area increases linearly in the flow direction (A_s is equal to perimeter times tube length).

The surface temperature for constant surface heat flux $\dot{q}_s$ is given by (Çengel and Ghajar 2015, Eq. (8–18))

$$\dot{q}_s = h(T_s - T_{\text{avg}}) \quad \Longrightarrow \quad T_s = T_{\text{avg}} + \frac{\dot{q}_s}{h}.$$

Also worth noting, is that the rate of change of the average temperature is equal the rate of change of the surface temperature in the flow direction (Çengel and Ghajar 2015, Eq.(8–22))

$$\frac{\mathrm{d}T_{\text{avg}}}{\mathrm{d}x} = \frac{\mathrm{d}T_s}{\mathrm{d}x} = \frac{\dot{q}_s p}{\dot{m} c_p} = \text{constant}.$$

Hence, for fully developed flus in a tube subject to constant surface heat flux, the temperature gradient is independent of x and therefore the shape of the temperature profile is invariant across the length of the tube (Çengel and Ghajar 2015, pp. 481–482).

6.3.2 Constant Surface Temperature

In the case of constant surface temperature, such as when phase change happens at the surface, the rate of heat transfer to or from the flowing fluid can be found using Newton's law of cooling (2)

$$\dot{Q} = hA_s \Delta T_{\text{avg}} = hA_s \left(T_s - T_{\text{avg}}\right)_{\text{avg}}.$$

The average fluid temperature at the tube exit is (Çengel and Ghajar 2015, Eq. (8–30))

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{hA_s}{\dot{m} c_p}\right).$$

Hence, the temperature difference between the fluid and the surface will decay exponentially in the flow direction.

The rate of heat transfer can also be expressed as (Çengel and Ghajar 2015, Eq. (8–32))

$$\dot{Q} = hA_s \Delta T_{1m},$$

where (Çengel and Ghajar 2015, Eq. (8–33))

$$\Delta T_{1m} = \frac{\Delta T_e - \Delta T_i}{\ln\left(\frac{\Delta T_e}{\Delta T_i}\right)}$$

is the *log mean temperature difference*, where $\Delta T_i = T_s - T_i$ and $\Delta T_e = T_s - T_e$ are the temperature differences between the surface and the fluid at the inlet and exit of the tube, respectively (Çengel and Ghajar 2015, pp. 482–484).

References

Çengel, Yunus A. and Afshin J. Ghajar (2015). *Heat and Mass Transfer: Fundamentals & Applications*. 5th edition. New York, NY: McGraw-Hill Education. 968 pp. ISBN: 978-0-07-339818-1.

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